

Create Table

```
CREATE TABLE Students (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Course VARCHAR(20),  
    Marks INT  
);
```

Create Log Table

```
CREATE TABLE Students_Log (  
    Student_ID INT,  
    Action VARCHAR(50)  
);
```

Insert Initial Records

```
INSERT INTO Students  
VALUES  
(101, 'Rahul', 'IT', 85),  
(102, 'Sneha', 'COMP', 90);
```

Display Table

```
SELECT * FROM Students;
```

1. BEFORE INSERT Trigger

(Prevents marks greater than 100)

DELIMITER //

```
CREATE TRIGGER Before_Insert_Trigger  
BEFORE INSERT  
ON Students  
FOR EACH ROW  
BEGIN  
    IF NEW.Marks > 100 THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT = 'Marks cannot be greater than 100';  
    END IF;  
END;  
//
```

DELIMITER ;

Correct Insert Data

```
INSERT INTO Students  
VALUES  
(103, 'Amit', 'IT', 78);
```

Incorrect Insert Data

```
INSERT INTO Students  
VALUES  
(104, 'Priya', 'COMP', 120);
```

2. AFTER INSERT Trigger

```
DELIMITER //  
  
CREATE TRIGGER After_Insert_Trigger  
AFTER INSERT  
ON Students  
FOR EACH ROW  
BEGIN  
    INSERT INTO Students_Log  
    VALUES (NEW.Student_ID, 'Record Inserted');  
END;  
//  
  
DELIMITER ;
```

Execute AFTER INSERT Trigger

```
INSERT INTO Students  
VALUES  
(105, 'Karan', 'COMP', 82);
```

Display Log Table

```
SELECT * FROM Students_Log;
```

3. BEFORE DELETE Trigger

```
DELIMITER //
```

```
CREATE TRIGGER Before_Delete_Trigger
BEFORE DELETE
ON Students
FOR EACH ROW
BEGIN
    IF OLD.Marks > 85 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'High Marks Record Cannot Be Deleted';
    END IF;
END;
//

DELIMITER ;
```

Correct Delete

```
DELETE FROM Students
WHERE Student_ID = 103;
```

Incorrect Delete

```
DELETE FROM Students
WHERE Student_ID = 102;
```

4. AFTER DELETE Trigger

```
DELIMITER //

CREATE TRIGGER After_Delete_Trigger
AFTER DELETE
ON Students
FOR EACH ROW
BEGIN
    INSERT INTO Students_Log
    VALUES (OLD.Student_ID, 'Record Deleted');
END;
//

DELIMITER ;
```

Execute AFTER DELETE Trigger

```
DELETE FROM Students
WHERE Student_ID = 105;
```

Display Log Table

```
SELECT * FROM Students_Log;
```

5. BEFORE UPDATE Trigger

```
DELIMITER //
```

```
CREATE TRIGGER Before_Update_Trigger
BEFORE UPDATE
ON Students
FOR EACH ROW
BEGIN
    IF NEW.Marks < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Marks cannot be negative';
    END IF;
END;
//
```

```
DELIMITER ;
```

Correct Update

```
UPDATE Students
SET Marks = 95
WHERE Student_ID = 101;
```

Incorrect Update

```
UPDATE Students
SET Marks = -10
WHERE Student_ID = 101;
```

6. AFTER UPDATE Trigger

```
DELIMITER //
```

```
CREATE TRIGGER After_Update_Trigger
AFTER UPDATE
ON Students
FOR EACH ROW
BEGIN
    INSERT INTO Students_Log
    VALUES (NEW.Student_ID, 'Record Updated');
END;
```

//

DELIMITER ;

Execute AFTER UPDATE Trigger

```
UPDATE Students  
SET Marks = 88  
WHERE Student_ID = 101;
```

Display Log Table

```
SELECT * FROM Students_Log;
```

Drop Triggers

```
DROP TRIGGER Before_Insert_Trigger;  
  
DROP TRIGGER After_Insert_Trigger;  
  
DROP TRIGGER Before_Delete_Trigger;  
  
DROP TRIGGER After_Delete_Trigger;  
  
DROP TRIGGER Before_Update_Trigger;  
  
DROP TRIGGER After_Update_Trigger;
```